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RESEARCH LABS

INNOVATION. AUTOMATION. ANALYTICS

PROJECT ON

Smart Food Ordering & Billing System

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BATCH NO : 493

ABOUT ME

- **Background:** B.Tech Student with interest in Python and Software Development
- **Why Data Science:** Passionate about solving real-world problems using data and automation
- **Experience:** Hands-on experience in Python projects and OOP-based application development
- **Profiles:**

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AGENDA

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- INTRODUCTION
- OBJECTIVE ON PROJECT
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- OOP CONCEPT USED
- DATA STRUCTURE USED
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- SAMPLE ORDER
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- SAMPLE CODE
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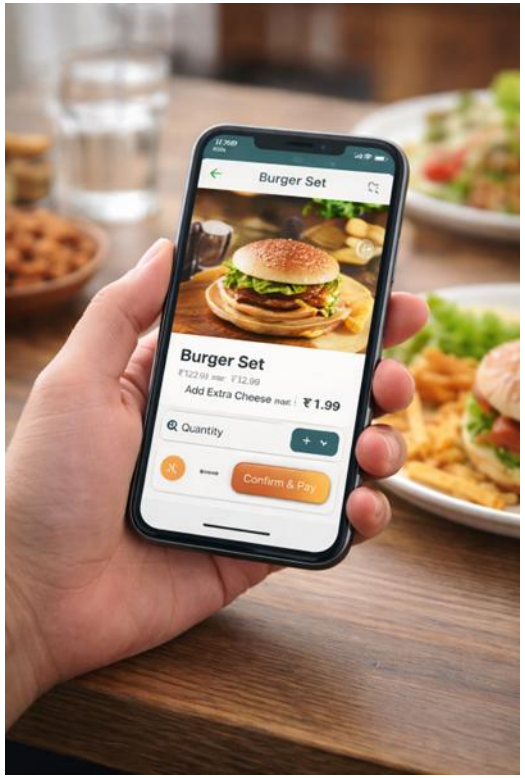

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BUSINESS STATEMENT

Manual restaurant billing systems:

- Are time-consuming
- Increase chances of human error
- Lack proper record management
- Do not provide structured invoices

The goal is to create a Python-based system that simplifies food ordering and billing.

INTRODUCTION

The **Smart Food Ordering & Billing System** is a console-based Python application developed to simulate a real-world restaurant billing system.

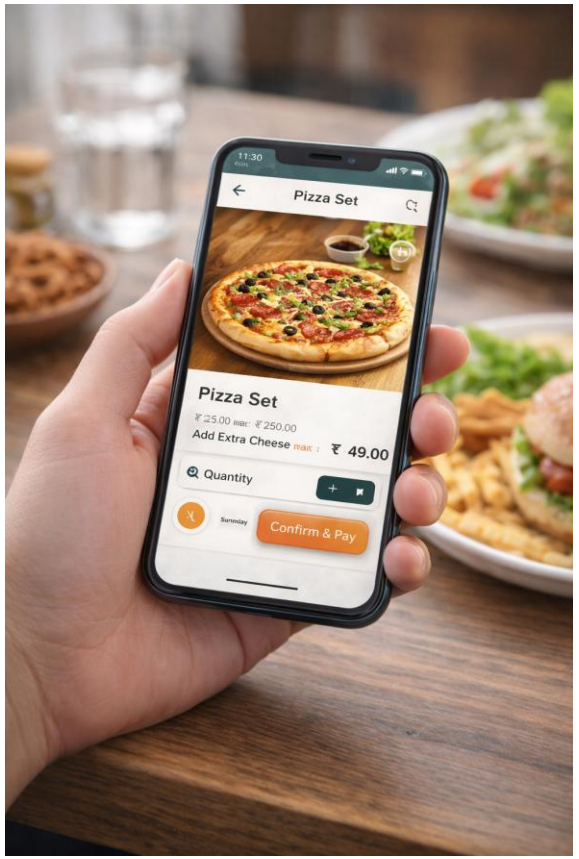
This system helps automate:

- Food ordering process
- Cart management
- Bill calculation with GST
- Invoice generation

It demonstrates how Python can be used to solve real-life business problems efficiently.

OBJECTIVE ON PROJECT

- To design a console-based food ordering system
- To implement cart functionality (add/remove items)
- To calculate total bill including GST
- To generate a formatted invoice
- To apply Object-Oriented Programming principles
- To demonstrate practical use of Python fundamentals



TECHNOLOGIES USED

Programming Language: Python

Concepts Used:

- Classes and Objects
- Functions
- Loops (while/for)
- Conditional Statements (if-else)
- Lists & Dictionaries
- String Formatting

SYSTEM ARCHITECTURE

Menu Module

- Stores food items and prices
- Displays available items

Cart Module

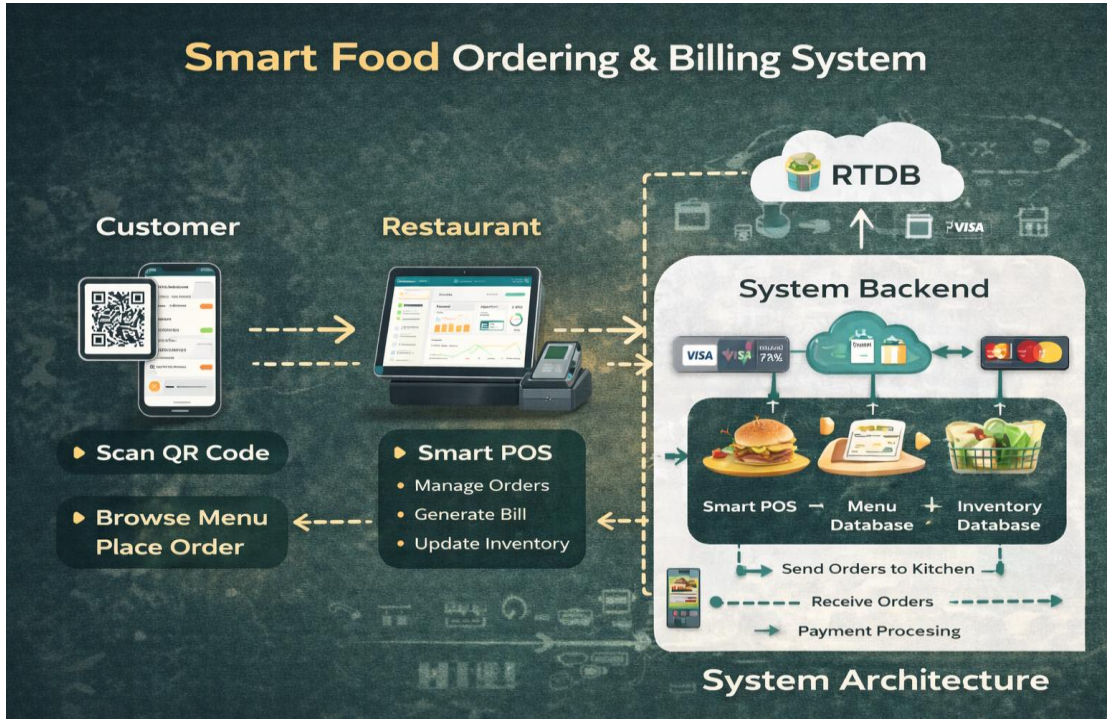
- Adds items
- Removes items
- Stores quantity

Billing Module

- Calculates subtotal
- Applies GST
- Generates total amount

Flow of System:

User → Menu → Cart → Billing → Invoice



OOP CONCEPT USED

Object-Oriented Programming plays a key role in this project.

Key OOP Concepts:

- Classes and Objects for room modeling
- Constructor (__init__) for initialization
- self keyword for instance variables
- Encapsulation of data and functions
- Using OOP makes the code:
- Reusable
- Organized
- Easy to expand

DATASTRUCTURE USED

DICTIONARY

Used to store:

- Food item names
- Prices

Example:

{"Burger": 100, "Pizza": 200}

LIST

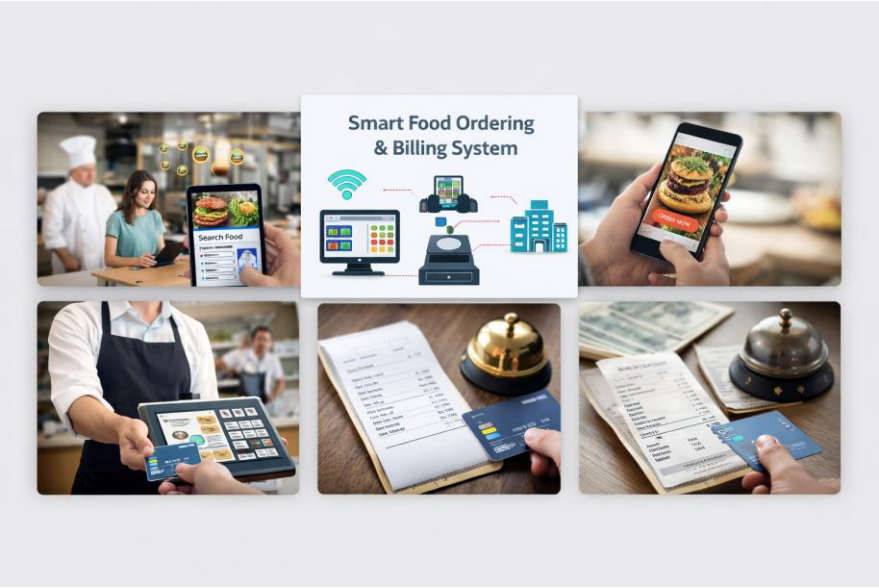
Used to:

- Store cart items
- Track selected food

.

WORKING PROCESS

Step 1: System displays menu
Step 2: User selects item number
Step 3: User enters quantity
Step 4: Item added to cart
Step 5: User can repeat or remove items
Step 6: System calculates subtotal
Step 7: GST added
Step 8: Final invoice displayed



SAMPLE ORDER

Example Order:
Burger – ₹120 × 2 = ₹240
Pizza – ₹250 × 1 = ₹250
Subtotal = ₹490
GST (5%) = ₹24.5
Total Bill = ₹514.5
Invoice shows:
Item name
Quantity
Price
GST
Grand Total

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RECOMENDATIONS

1.Mobile Application Integration

Develop a **mobile app (Android / iOS)** so customers can easily order food directly from their smartphones

2.Online Payment Integration

Add secure payment gateways like **UPI, Debit Card, Credit Card, and Wallets** for faster transactions.

3.Real-Time Order Tracking

Customers should be able to **track the status of their order** (Preparing → Cooking → Ready → Delivered).

4.AI-Based Food Recommendations

Use **Artificial Intelligence** to suggest food items based on customer preferences and previous orders.

5.Inventory Management System

Automatically update **ingredient stock** when an order is placed to help restaurants manage inventory efficiently.

SAMPLE CODE

```python
class Order:
    def __init__(self):
        self.cart = []
        self.total = 0

    def add_to_cart(self, item_name, price, quantity):
        cost = price * quantity
        self.cart.append({
            "item": item_name,
            "price": price,
            "quantity": quantity,
            "cost": cost
        })
        self.total += cost
        print(f"{item_name} added to cart successfully!")
```

```python
order = Order()
order.add_to_cart("Burger", 120, 2)
order.add_to_cart("Pizza", 250, 1)

print(order.cart)
print("Total:", order.total)
```

OUTPUT

```
=========== ORDER SYSTEM ===========
1. View Menu
2. Add Item
3. Remove Item
4. View Cart
5. Print Bill
6. Clear Cart
7. Exit
====================================
Enter your choice:  1

========== MENU ==========
1. Burger      : ₹120
2. Fries       : ₹80
3. Coke        : ₹50
4. Pizza       : ₹200
5. Wrap        : ₹150
==========================
```

```
=========== ORDER SYSTEM ===========
1. View Menu
2. Add Item
3. Remove Item
4. View Cart
5. Print Bill
6. Clear Cart
7. Exit
====================================

Enter your choice:  4


========== YOUR CART ==========
Burger      x2 = ₹240
Wrap        x1 = ₹150
--------------------------------
Subtotal: ₹390
================================
```

```
=========== FINAL BILL ===========
Burger      x2 = ₹240
Wrap        x1 = ₹150
----------------------------------
Subtotal      : ₹390.00
GST (5%)      : ₹19.50
Grand Total   : ₹409.50
==================================
```

```
Enter your choice:  6
Cart cleared successfully.

=========== ORDER SYSTEM ===========
1. View Menu
2. Add Item
3. Remove Item
4. View Cart
5. Print Bill
6. Clear Cart
7. Exit
====================================
Enter your choice:  7
Thank you for using Burger Ordering System.
```

CONCLUSION

The Smart Food Ordering & Billing System:

- Successfully simulates a restaurant billing process
- Demonstrates practical use of Python fundamentals
- Applies Object-Oriented Programming concepts
- Provides a structured and automated billing solution

This project enhances problem-solving skills and real-world software development understanding.

Q&A

THANK YOU

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